STPM PHYSICS · GEOMETRICAL OPTICS

# SIGN CONVENTION REMINDER

## REAL IS POSITIVE

Real object, real image, real focus → **+**

Virtual object, virtual image, virtual focus → **−**

**Ask yourself one question:**

## "Where does the light actually go?"

### MIRROR

Light bounces **back**.

**Real side = IN FRONT**

### LENS & CURVED SURFACE

Light goes **through**.

**Real image side = THE OTHER SIDE**

**3 steps, every question**

1. Put the sign on every known value ($u$, $f$, $r$) **before** you substitute.
2. Solve for $v$. Do not drop the minus sign.
3. Read the sign: **+ real**, **− virtual**. Then **say where** the image is.

### REAL IMAGE

Inverted · can be caught on a screen

### VIRTUAL IMAGE

Upright · cannot be caught on a screen

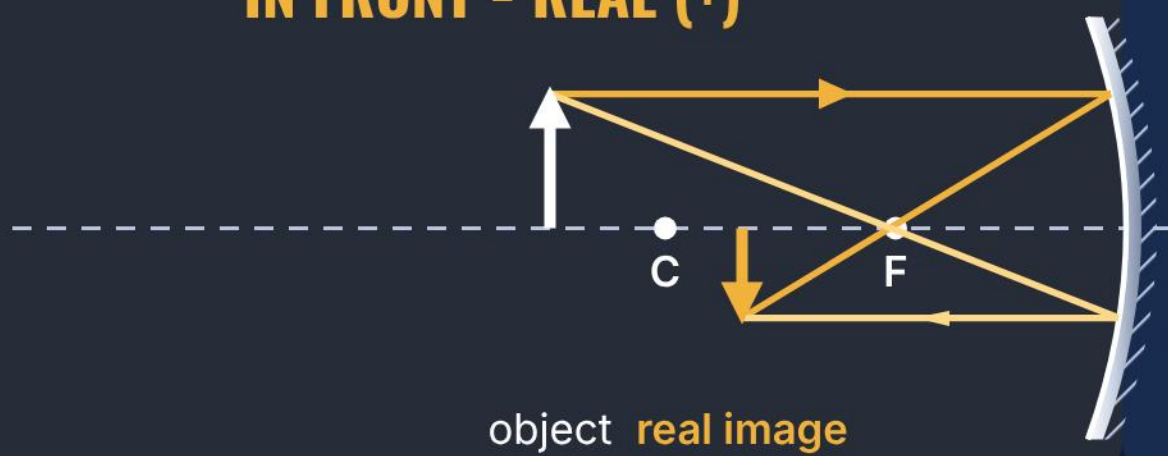
Size: magnification $\frac{|v|}{|u|}$ > 1 means magnified, < 1 means diminished.

REAL IS POSITIVE

# MIRRORS

$$\frac{1}{u}+\frac{1}{v}=\frac{1}{f} \qquad f=\frac{r}{2}$$

## IN FRONT = REAL (+)




object **real image**

## BEHIND = VIRTUAL (−)

silvered back:
no real light here

Image **in front** of mirror

**$v$ = +**

Real · inverted

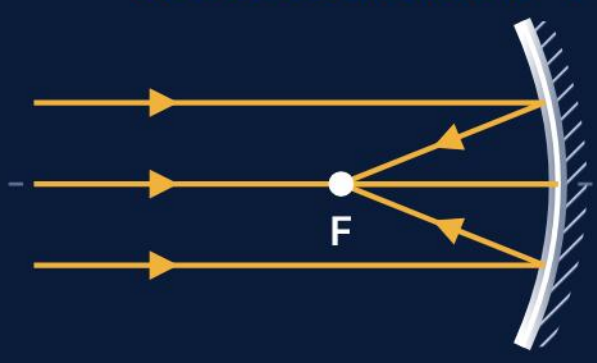
Image **behind** mirror

**$v$ = −**

Virtual · upright

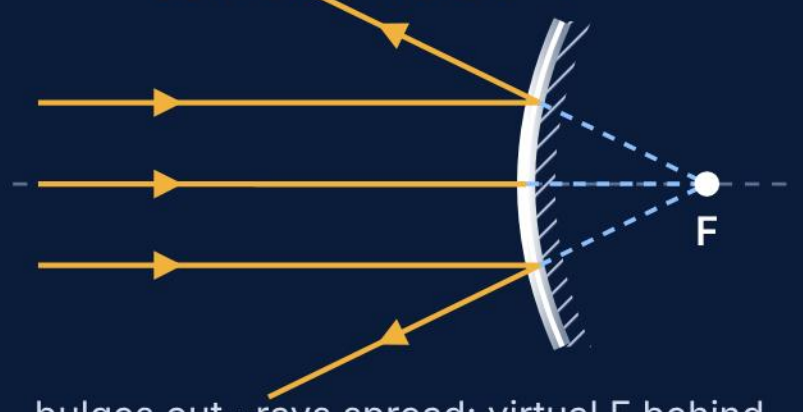
## Know the shapes: concave vs convex

### CONCAVE MIRROR · f = +




aves in (like inside of a spoon) · rays meet: real F

### CONVEX MIRROR · f = −




bulges out · rays spread: virtual F behind

## Quick tips

- **Convex mirror** (e.g. car side mirror): image is always virtual, upright, diminished, so $v$ is always −.
- **Concave mirror** with object inside F ($u < f$): image is virtual and magnified behind the mirror (e.g. make-up mirror).

## EXAMPLE

Concave mirror, $f$ = +10 cm, object at $u$ = +5 cm.

$$\frac{1}{v}=\frac{1}{10}-\frac{1}{5}=-\frac{1}{10} \quad \rightarrow \boldsymbol{v = -10\ \text{cm}}$$

**Minus sign: virtual, upright image 10 cm behind the mirror.**

REAL IS POSITIVE

# LENSES

$$\frac{1}{u}+\frac{1}{v}=\frac{1}{f}$$

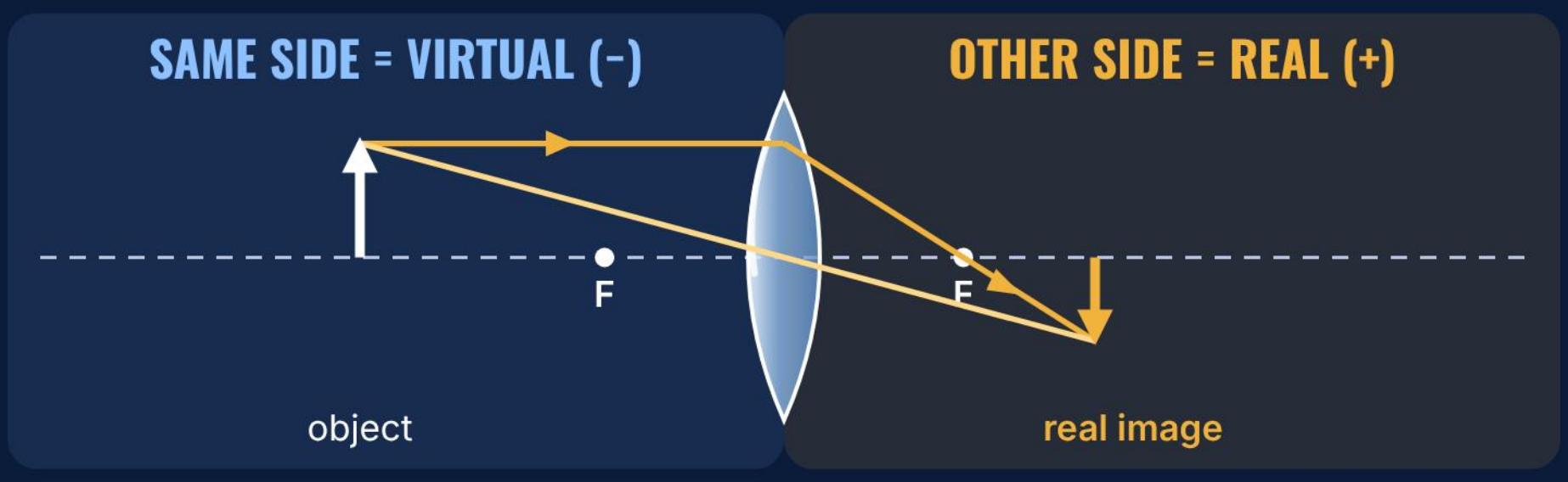



Image on the **other side** of the lens

**$v$ = +**

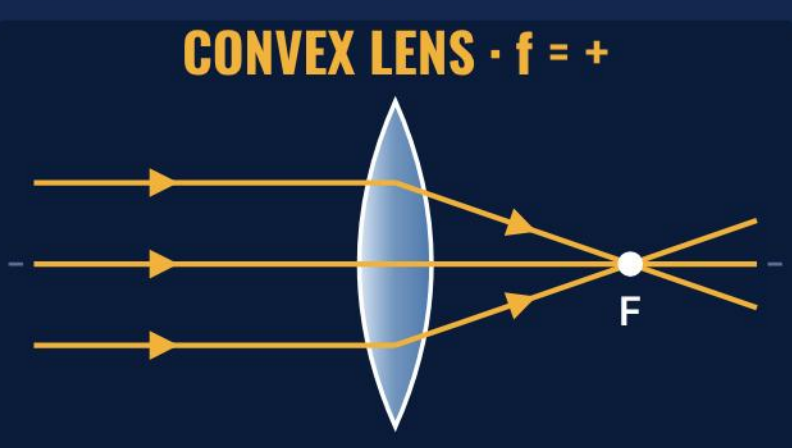
**Real · inverted**

Image on the **same side** as the object

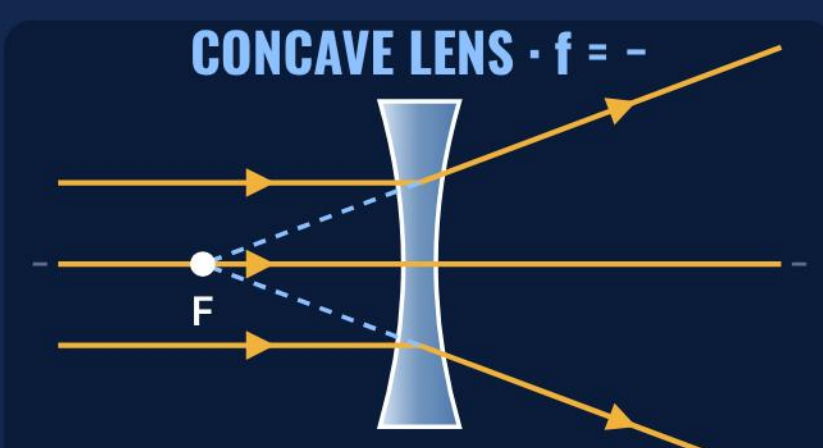
**$v$ = −**

**Virtual · upright**

## Know the shapes: convex vs concave




thick in the middle · rays meet: real F




thin in the middle · rays spread: virtual F

## Lensmaker's equation

$$\frac{1}{f}=(n-1)\left(\frac{1}{r_1}-\frac{1}{r_2}\right)$$

$r$ is **+** if its centre of curvature is on the **outgoing** side, **−** if on the **incoming** side (light travels left → right).



**Biconvex**
$r_1 + r_2 -$

**Plano-convex**
$r_1 + r_2 = \infty$

**Biconcave**
$r_1 - r_2 +$

**Plano-concave**
$r_1 - r_2 = \infty$

## Quick tips

- **Concave lens** with a real object: image is always virtual, upright, diminished, so $v$ is always −.
- **Convex lens** with object inside F ($u < f$): virtual, magnified image (magnifying glass), so $v$ is −.

## EXAMPLE

Convex lens, $f$ = +10 cm, object at $u$ = +15 cm.

$$\frac{1}{v}=\frac{1}{10}-\frac{1}{15}=\frac{1}{30} \rightarrow \boldsymbol{v = +30\ \text{cm}}$$

**Plus sign: real, inverted image 30 cm on the other side of the lens.**

REAL IS POSITIVE

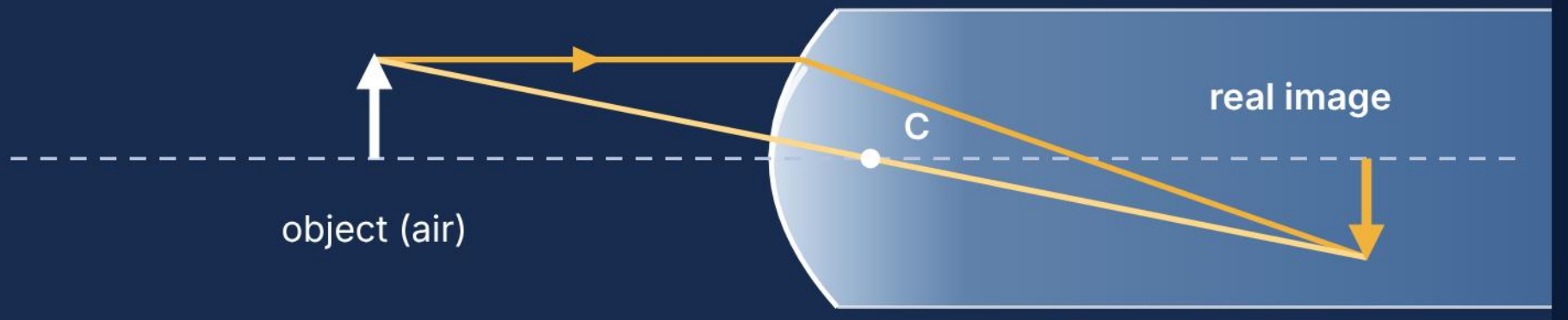
# REFRACTION AT A CURVED SURFACE

$$\frac{n_1}{u} + \frac{n_2}{v} = \frac{n_2 - n_1}{r}$$

$n_1$ = medium where the **object** is

$n_2$ = medium the light **enters**

**MEDIUM 1 ($n_1$) = VIRTUAL (−)** **MEDIUM 2 ($n_2$) = REAL (+)**




glass rod · C in medium 2 → r = +

Image in **medium 2** (after the surface)

***v* = +**

**Real image**

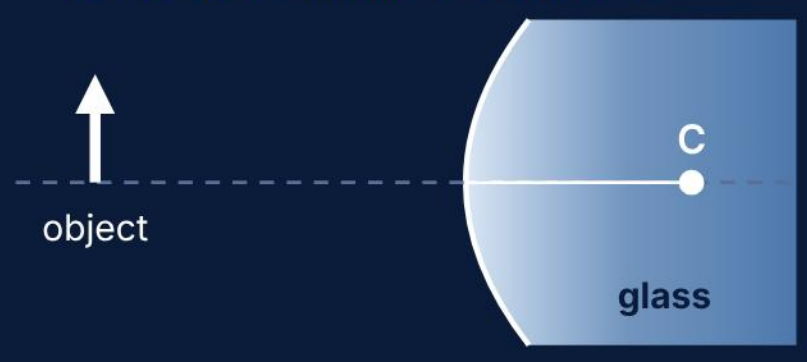
Image in **medium 1** (same side as object)

***v* = −**

**Virtual image**

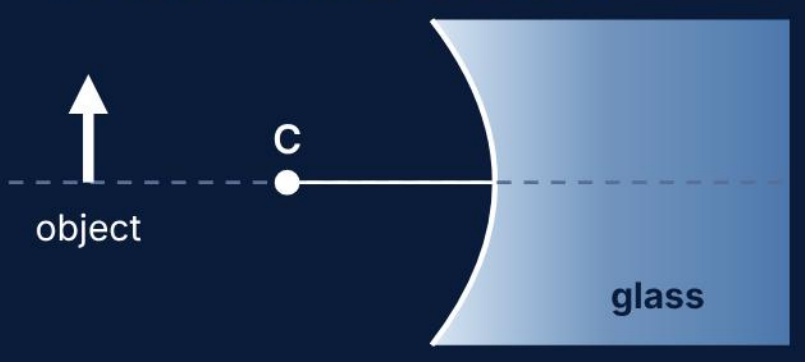
## Radius sign: stand at the object and look at the surface

**CONVEX TOWARDS OBJECT · r = +**




surface bulges towards the object
C is in medium 2 (glass)

**CONCAVE TOWARDS OBJECT · r = −**




surface caves away from the object
C is in medium 1 (object side)

## Quick tips

- $n_1$ is always where the light **comes from**. Light going glass → air: $n_1$ = glass, $n_2$ = air.
- Plane surface: $r = \infty$, so $\frac{n_1}{u} + \frac{n_2}{v} = 0$ (apparent depth; $v$ comes out −).

## EXAMPLE

Object in air, 8 cm from the convex end of a glass rod ($n$ = 1.5, $r$ = +2 cm).

$$\frac{1}{8} + \frac{1.5}{v} = \frac{1.5 - 1}{2} \rightarrow \frac{1.5}{v} = 0.125 \rightarrow \boldsymbol{v = +12\ \text{cm}}$$

**Plus sign: real image 12 cm inside the glass.**

REFRACTION AT A CURVED SURFACE

# WHICH WAY IS THE LIGHT GOING?

**Same equation for both directions:**

$$\frac{n_1}{u} + \frac{n_2}{v} = \frac{n_2 - n_1}{r}$$

$n_1$ = where light comes FROM · $n_2$ = where light GOES

## LESS DENSE → MORE DENSE

e.g. air → glass, air → water

$n_1 < n_2$

**$(n_2 - n_1)$ is +**

Light bends **towards** the normal.

## MORE DENSE → LESS DENSE

e.g. glass → air, fish in a bowl

$n_1 > n_2$

**$(n_2 - n_1)$ is –**

Light bends **away** from the normal.

**Do not flip any signs yourself.** The direction of light only decides which $n$ is $n_1$. Signs of $u$, $v$ and $r$ follow the same rules as card 4.

## EXAMPLE 1 · LESS → MORE DENSE

Object in air, 8 cm from the convex end of a glass rod ($n$ = 1.5). Surface bulges towards the object: $r$ = +2 cm.

$$\frac{1}{8} + \frac{1.5}{v} = \frac{1.5 - 1}{+2} \rightarrow \boldsymbol{v = +12\ \text{cm}}$$

**Real image, 12 cm inside the glass.**

## EXAMPLE 2 · MORE → LESS DENSE

Fish 10 cm behind the front of a spherical bowl (radius 15 cm, $n = \frac{4}{3}$), seen from outside. From the fish, the surface caves away: $r$ = −15 cm.

$$\frac{4/3}{10} + \frac{1}{v} = \frac{1 - 4/3}{-15} = \frac{1}{45} \rightarrow \boldsymbol{v = -9\ \text{cm}}$$

**Virtual image, 9 cm behind the glass (inside the water). The fish looks closer.**

REFRACTION AT A CURVED SURFACE

# THREE MEDIA? ONE SURFACE AT A TIME

1. Surface 1: use its own $n_1$ (from) and $n_2$ (to). Find $v_1$.
2. The image from surface 1 becomes the **object** for surface 2. Measure $u_2$ from surface 2.
3. Surface 2: new $n_1$ and $n_2$, solve again for $v_2$.

**REAL OBJECT · $u_2$ = +**

Image 1 is **before** surface 2. Light spreads out from it.

**VIRTUAL OBJECT · $u_2$ = −**

Image 1 would form **beyond** surface 2. Light is still converging.

## EXAMPLE 1 · LESS DENSE → DENSER

**Air (1) → water ($\frac{4}{3}$) → glass ($\frac{3}{2}$).** Object in air, 60 cm from a convex air–water surface ($r$ = +10 cm). The water is 40 cm thick, then a flat water–glass boundary.

**Surface 1 (air → water)**

$$\frac{1}{60} + \frac{4/3}{v_1} = \frac{4/3 - 1}{+10} \rightarrow \mathbf{v_1 = +80\ cm}$$

80 cm is beyond the glass (only 40 cm away), so it is a **virtual object**: $\mathbf{u_2 = -(80 - 40) = -40\ cm}$.

**Surface 2 (water → glass, flat: $r = \infty$)**

$$\frac{4/3}{-40} + \frac{3/2}{v_2} = 0 \rightarrow \mathbf{v_2 = +45\ cm}$$

**Real image, 45 cm inside the glass.**

## EXAMPLE 2 · DENSER → LESS DENSE

**Glass ($\frac{3}{2}$) → water ($\frac{4}{3}$) → air (1).** A coin under a 6 cm glass slab, with 8 cm of water above the slab. Looking down from air.

**Surface 1 (glass → water, flat)**

$$\frac{3/2}{6} + \frac{4/3}{v_1} = 0 \rightarrow \mathbf{v_1 = -\frac{16}{3}\ cm}$$

Image 1 is 5.33 cm below the slab top, before the water surface, so it is a **real object**: $\mathbf{u_2 = 8 + \frac{16}{3} = \frac{40}{3}\ cm}$.

**Surface 2 (water → air, flat)**

$$\frac{4/3}{40/3} + \frac{1}{v_2} = 0 \rightarrow \mathbf{v_2 = -10\ cm}$$

**Virtual image: the coin looks 10 cm below the water surface (real depth 14 cm).**

**Check for flat layers:** apparent depth = $\frac{t_1}{n_1} + \frac{t_2}{n_2} = \frac{6}{3/2} + \frac{8}{4/3} = 4 + 6$ = 10 cm ✓